

Please upload your completed assignment AND any R Code used to complete the work on Moodle.

1. In this question we will work with the tidyverse package, as well as Lego dataset in Moodle.

- a. Load the tidyverse package.**
- b. Import the lego_sales dataset.**
- c. What is the data structure of the lego_sales dataset?**
- d. For each data type below list all the variables that are that type:**
 - i. Numeric: us_price
 - ii. Logical: n/a
 - iii. Character: first_name, last_name, phone_number, number, theme, subtheme, name, image_url
 - iv. Integer: age, set_id, year, pieces, quantity

Let's pretend you are a data analyst for Lego. At your weekly team meeting your boss has a new project for you. They want to know which Lego theme generates the most total revenue, based on price and quantity sold. They tell you to define revenue as price times quantity sold.

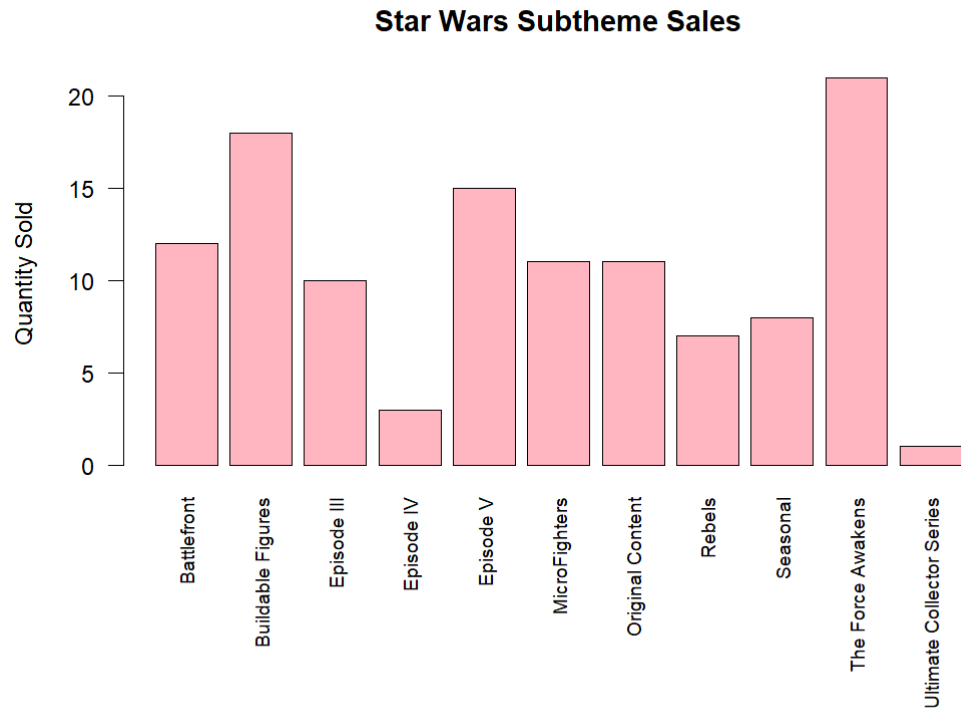
- e. What type of question (based on the 6 discussed in lecture) is your boss asking you to answer?**

The type of question my boss is asking me to answer is a descriptive question because he wants me to summarise the total revenue for each Lego theme and identify which one is the highest

- f. What Lego theme generates the most revenue?**

Star Wars

- g. Using the theme identified in part f, create an appropriate visual that shows how many products of each subtheme were bought.



- h. At your next weekly team meeting your boss calls on you to present what you have found from the dataset. Write out what you would say to the team.

I analysed the Lego sales dataset and calculated the total revenue for each theme by multiplying the price of the Lego and the quantity sold. My findings show that the Star Wars theme generate the highest total revenue which was \$4,448.

I also examined the distribution of sales within the star wars theme and found that some subthemes contribute to the total revenue more heavily than others. This suggests that certain product lines within the Star Wars theme are particularly popular and might be worth focusing on for future marketing or production decisions.

2. For this question use the gapminder dataset found in the gapminder package. Pretend you are working for a non-governmental organisation that is trying to determine which region to focus their efforts on in order to increase life expectancy.

- a. What is the range of years represented in the dataset?

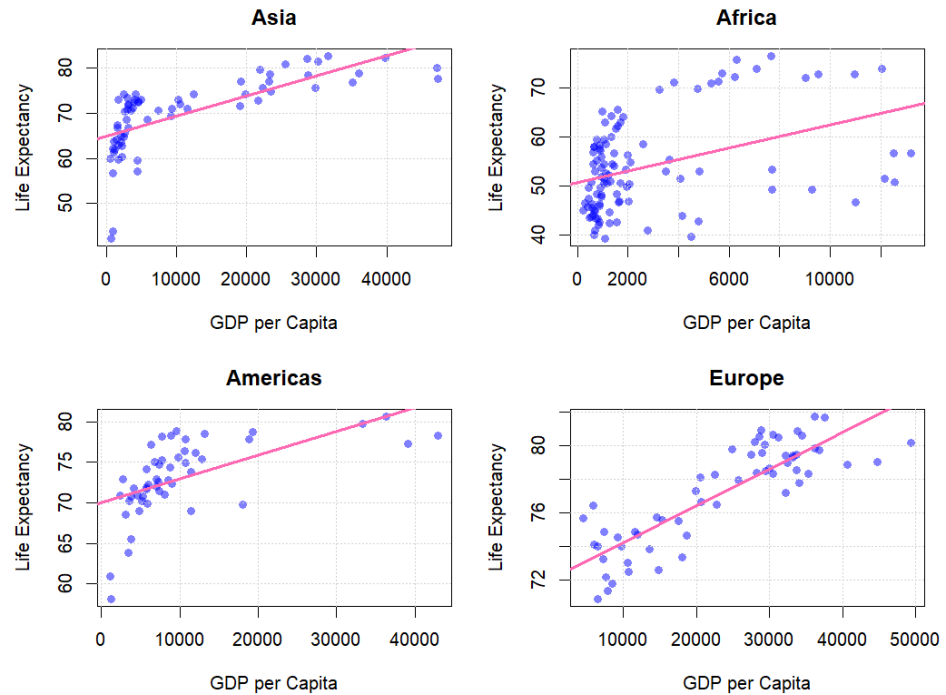
1952-2007

We want to base our recommendations on the most up-to-date data possible. Please only use available data from the years 2000-2026.

- b. Fill in the following table using the dataset:

Continent	Mean life Expectancy	Maximum GDP Per Capita	Total Population
Africa	54.1	13206	1,763,263,608
Americas	73.0	42952	1,748,643,946
Asia	70.0	47307	7,413,756,030
Europe	77.2	49357	1,164,322,398
Oceania	80.2	34435	48,004,776

- c. Using the *four continents* that have the highest total population create a visual (or multiple visuals arranged in a grid) that show how life expectancy changes with changes in GDP per Capita. Make sure the axes have labels that are clear.



- d. **Your team lead has asked you to present your findings and recommendations to the team. Do all 4 continents have the same trends in life expectancy as a function of GDP per Capita? What continent would you recommend focusing efforts on? Are there any other observations you would make sure to mention?**

Based on the four continent scatterplots, we see clear differences in how GDP relates to life expectancy. All four continents show a positive relationship, but not the same trend: Europe and the Americas rise quickly and then level off at high GDP, Asia shows strong gains across the range, and Africa has both the lowest life expectancy and the weakest improvement per unit of GDP. For example, at GDP=2000, Africa's life expectancy is around 62 years while Asia is already near 72.

Therefore, I recommend directing resources toward Africa, focusing on why economic growth there translates into smaller life-expectancy gains and identifying non-economic interventions that could accelerate progress. I would also note the plateau in high-GDP regions and the larger variability within Africa and Asia, which suggests uneven development within continents.

- 3. Pretend you are working as a data analyst at a hospital. The hospital recently introduced a new discharge support program for patients recovering from minor surgery. The program includes extra instructions, follow-up phone calls, and additional recovery resources.**

Hospital administrators want to know whether patients who participate in the new program recover faster than those who receive the usual discharge process. Here are the top rows of the dataset you are shown:

	A	B	C	D	E	F	G	
1	Patient ID	Surgery Code	DOB	Program	Sx Date	Discharge Date	RTN Date	
2	1	11190	12/03/1994	R	18/01/2024	20/01/2024	30/01/2024	
3	2	42495	28/07/1987	N	27/02/2024	28/02/2024	2/03/2024	
4	3	32659	3/11/2001	R	9/03/2024	11/03/2024	12/03/2024	
5	4	69554	19/01/1999	R	22/04/2024	24/04/2024	2/05/2024	
6	5	51884	6/05/1992	N	11/06/2024	13/06/2024	16/06/2024	
7	6	49395	14/09/2000	R	30/07/2024	31/07/2024	7/08/2024	
8	7	40005	2/02/1985	R	5/09/2024	7/09/2024	8/09/2024	
9	8	11192	21/08/1997	R	19/10/2024	20/10/2024	28/10/2024	
10	9	26706	30/12/1990	R	2/12/2024	5/12/2024	12/12/2024	
11	10	51893	11/04/2003	N	14/01/2025	17/01/2025	18/01/2025	
12	11	40932	9/10/1996	N	26/02/2025	27/02/2025	9/03/2025	
13	12	32659	25/06/1989	N	17/03/2025	19/03/2025	3/04/2025	
14	13	40018	3/03/1998	R	8/05/2025	11/05/2025	18/05/2025	
15	14	11186	7/01/1993	N	21/06/2025	23/06/2025	30/06/2025	
16	15	14587	1/07/2002	N	13/08/2025	16/08/2025	26/08/2025	
17	16	34134	18/11/1995	R	4/10/2025	5/10/2025	11/10/2025	
18	17	65440	27/05/1988	R	29/11/2025	2/12/2025	8/12/2025	
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a. What questions would you ask during the meeting to clarify the task?

1. How is recovery time measured?
2. Are patients randomly assigned to the program?
3. Are there differences between patients who choose the program and those who do not?
4. What other variables are recorded (e.g., age, health status)?
5. Are there missing or incomplete observations?

From the questions and conversations you had in part a you learn the following additional information:

- Recovery time is measured as the number of days until a patient reports returning to normal daily activities.
- Patients could choose whether to enroll in the new program.

- **Some doctors encouraged patients they thought might need extra support to join the program.**
- **Patients in the program tend to be slightly younger on average.**
- **The hospital recorded the date a patient reported returning to normal activities (RTN Date).**
- **The hospital director asks: “Did the new discharge program help patients recover faster?”**

a. What type of question/questions (based on the 6 discussed in lecture) do the hospital administrators want you to answer?

This is a causal question because the hospital wants to know whether the program causes faster recovery times.

b. Outline the steps you would take after the meeting to complete the analysis. (You do not need to do any coding or actually carry out the steps, just walk me through the steps you would take.)

- I would clearly define the statistical question based on the domain problem
- Import and clean the dataset
- Explore the data using summaries and visualisations
- Compare recovery times between program and non-program groups
- Check assumptions (normality, independence, outliers)
- Perform an appropriate statistical test (e.g., two-sample t-test)
- Consider potential confounding variables (e.g., age differences)
- Interpret results in context
- Report findings and limitations